



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SDCMPU LTD.

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

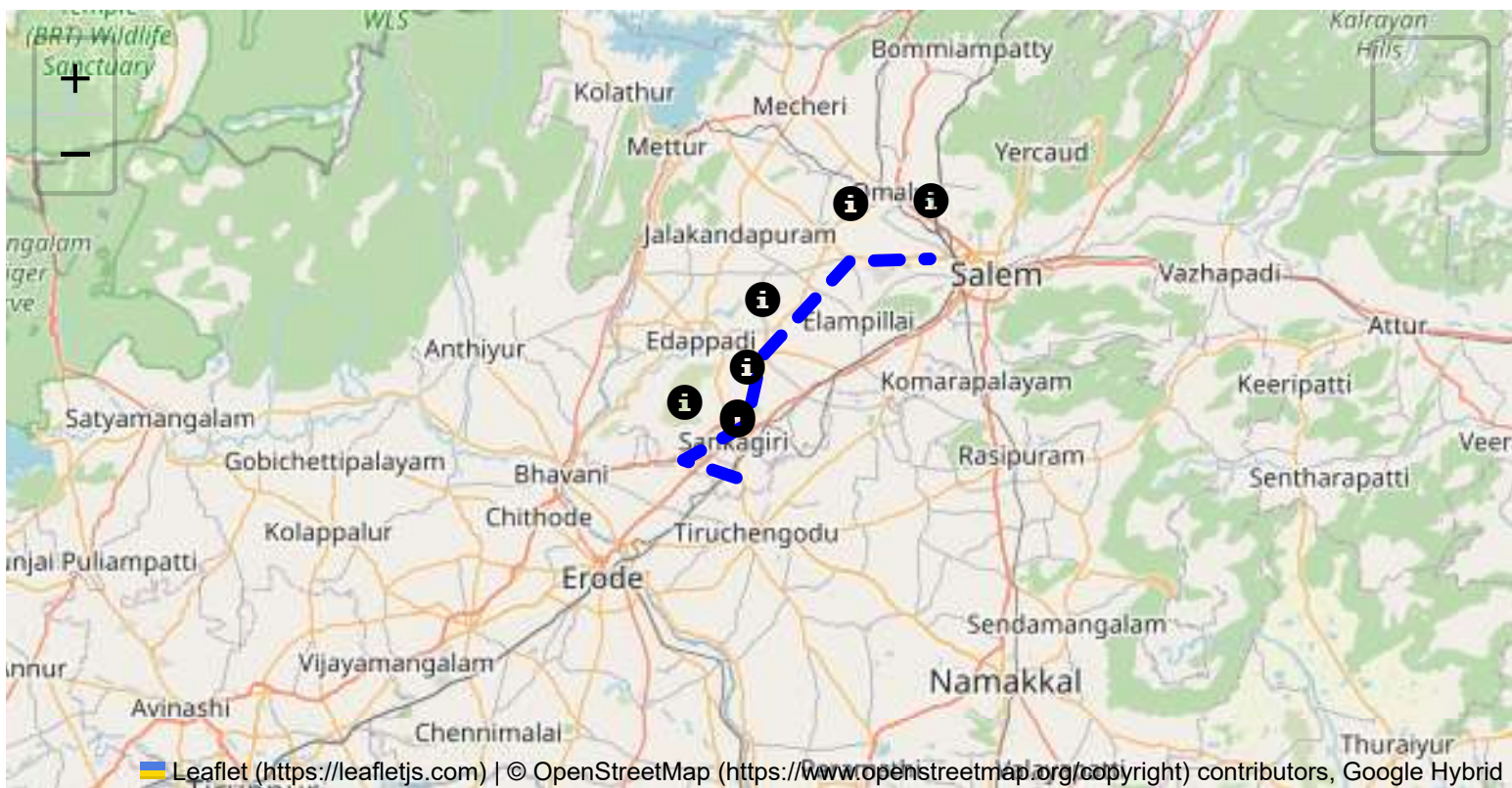
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

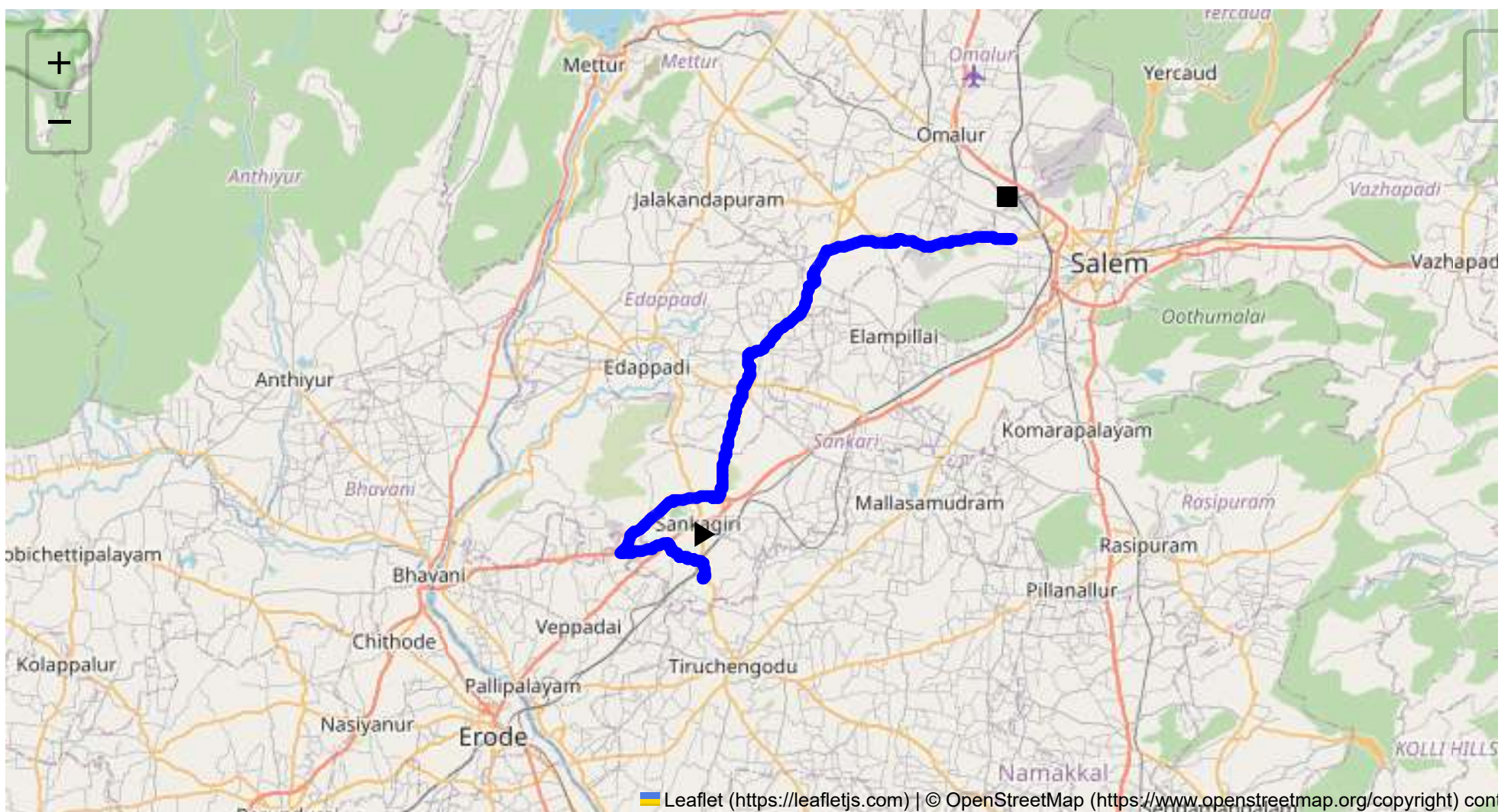
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 54.69 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (11.670288, 78.085531)



Welcome to the Journey Risk Management Study

Route Overview

The route covers a distance of approximately 54.69 kilometers and connects Sangagiri to Thalavaipatti in Salem District, Tamil Nadu. En route, it passes through several key locations including Puthur, Padaiveedu, and Erumaipatti. The journey involves traveling on regional roads and major highways, notably the Bhavani Main Road and Salem-Kochi Highway.

Weather Conditions

Typical weather in this region includes:

- **Monsoon Season (June to September):** Heavy rainfall can cause road flooding and landslides, potentially obstructing travel.
- **Summer (March to June):** High temperatures up to 40°C can lead to road surface degradation.
- **Winter (November to February):** Mild conditions, generally safe for travel with occasional fog in the morning.

Traffic Patterns

- **Peak Hours:** Mornings (8:00 AM - 10:00 AM) and evenings (5:00 PM - 7:00 PM) experience higher traffic volumes, particularly near urban areas like Sangagiri and Salem.
- **Congestion-Prone Areas:** Salem-Kochi Highway can witness congestion due to its role as a major transport route. The approach to Salem is particularly busy.

Road Quality and Infrastructure

- **Road Quality:** Varies from well-maintained highways to rural roads that might be narrow and less well-maintained.
- **Infrastructure:** Adequate facilities in urban areas but limited petrol stations and rest stops in rural stretches.

Alternative Routes for Emergencies

- **Alternative Route:** From Sangagiri, consider taking the NH544 towards Erode and then shifting towards Salem via smaller feeders if there are roadblocks or severe congestion on the primary route.

Local Regulations

- **Hazardous Material Transport:** Requires permits and adherence to designated routes. Restrictions may apply within city limits, especially during peak hours.

Historical Incidents

- **Reported Incidents:** Occasional accidents involving trucks due to narrow road conditions and sharp turns, particularly near hilly terrains.

Environmental Considerations

- **Sensitive Areas:** The route traverses agricultural zones; any spillage of hazardous materials could have significant environmental impacts. Noise pollution near populated areas must be minimized.

Communication Coverage

- **Coverage:** Generally good in urban and semi-urban areas but may experience dead zones in rural stretches, particularly near Erumaipatti.

Emergency Response Times

- **Urban Areas (Sangagiri, Salem):** Approximately 20-30 minutes due to proximity to emergency services.
- **Rural Areas (Puthur, Padaiveedu):** Can exceed 45 minutes depending on the location and time.

Overall Summary of Risk Assessment

- **Primary Risks:** Include weather-related disruptions, traffic congestion, and road infrastructure quality.
- **Recommendation:** Drivers should be prepared for changing road and weather conditions and always have emergency provisions. Regular communication checks are recommended, especially when approaching known dead zones. Adhering to local regulations is crucial for the safe transport of hazardous materials. Alternative route planning is advised to handle unexpected delays or obstructions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.6709299, 78.0889282	10 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44898, 77.87410	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46048, 77.85036	30 KM/Hr
9	Turn	Medium	11.46303, 77.84945	30 KM/Hr
10	Turn	Medium	11.46318, 77.84913	30 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Turn	High	11.45631, 77.81668	15 KM/Hr
13	Turn	High	11.49409, 77.88004	15 KM/Hr
14	Turn	Medium	11.49419, 77.88015	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	High	11.49383, 77.88442	15 KM/Hr
16	Turn	Medium	11.49419, 77.88454	30 KM/Hr
17	Turn	High	11.49448, 77.88490	15 KM/Hr
18	Turn	Medium	11.56083, 77.89921	30 KM/Hr
19	Turn	Medium	11.57048, 77.90224	30 KM/Hr
20	Turn	Medium	11.58304, 77.90656	30 KM/Hr
21	Turn	Medium	11.63840, 77.94820	30 KM/Hr
22	Turn	High	11.66254, 77.96043	15 KM/Hr
23	Turn	High	11.66790, 78.00500	15 KM/Hr
24	Turn	High	11.66946, 78.00537	15 KM/Hr
25	Turn	High	11.66903, 78.00811	15 KM/Hr
26	Turn	High	11.66951, 78.00827	15 KM/Hr
27	Blind Spot	Blind Spot	11.66936, 78.00887	10 KM/Hr
28	Turn	High	11.67168, 78.00922	15 KM/Hr
29	Blind Spot	Blind Spot	11.67093, 78.08893	10 KM/Hr
30	Turn	High	11.67087, 78.08893	15 KM/Hr
31	Turn	High	11.67071, 78.08567	15 KM/Hr
32	Turn	High	11.67041, 78.08559	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
3	hospital	ESI Hospital, Salem	11.671506, 78.0818924	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
2	college	Salem Mohan Kumaramangalam Medical College	11.6692169, 78.0671082	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.6709299, 78.0889282



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49419, 77.88015



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88454



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49448, 77.88490



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57048, 77.90224



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58304, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63840, 77.94820



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66254, 77.96043



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66790, 78.00500



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.66936, 78.00887



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67168, 78.00922



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.67093, 78.08893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67087, 78.08893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67071, 78.08567



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67041, 78.08559
